



Internal and international migration of scholars worldwide: Trends, patterns, and inter-relationships

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- Theoretical impressions
 - Internal versus international migration (1, 2)

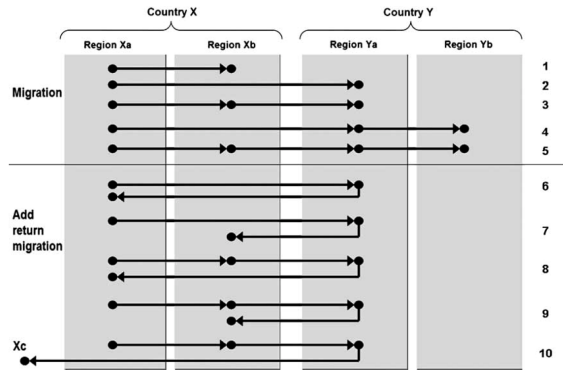
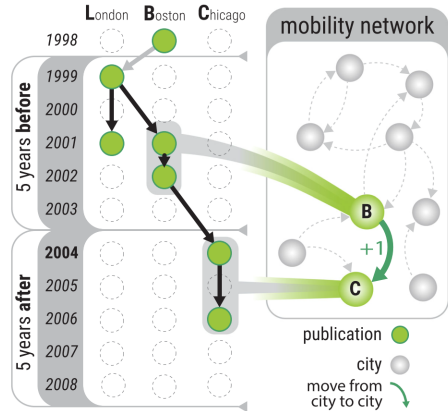


Figure 1. Migration pathways

(King, Skeldon (2010))

- ▶ Theoretical impressions
 - ▶ Internal versus international migration (1, 2)
 - ▶ Research Trails (3)
- ▶ Methodological impressions
 - ▶ Using bibliometric data alongside demographic life events (4)
 - ▶ Repurposing bibliometric data for knowledge transfer (5, 6)
 - ▶ Repurposing bibliometric data as network of scientific mobility among cities (7)



(Verginer and Riccaboni (2020))

Introduction



- Theoretical impressions
 - Internal versus international migration (1, 2)
 - Research Trails (3)
- Methodological impressions
 - Using bibliometric data alongside demographic life events (4)
 - Repurposing bibliometric data for knowledge transfer (5, 6)
 - Repurposing bibliometric data as network of scientific mobility among cities (7)
 - Repurposing bibliometric data for migration research (8, 9)

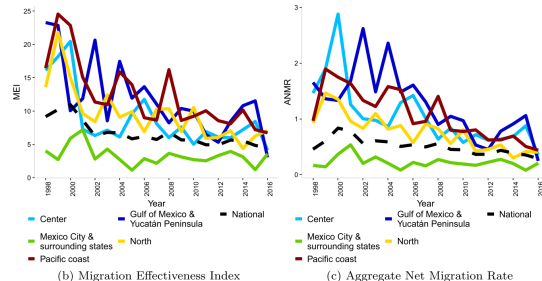


Figure 8 Migration intensity indices, over time Dashed line represent country indices

(Miranda Gonzalez et al. (2020))

Our proposal

Repurposing bibliometric data for:

Internal versus **international** migration of **scholars** worldwide
(10)

A simple and scalable idea:

Changes in **institutional affiliations** can be used to infer changes in **residence over time** for individual scholars and for populations

Our questions

1. Which are the (persistently) **attractive** sending/receiving regions internally/internationally?
2. Is academic **talent circulation** happening more *within* or *between* countries?
3. What are the **mutual effects** of these two systems of migration?

- **Scopus 1996-2020** article/review publications (**28+** million, **8+** million scholars)
- Provided by German Competence Network for Bibliometrics via Max Planck Digital Library
- Encoding **country** of affiliation from bibliometric record (not perfect, need proper disambiguation)
- **Organization name disambiguation** using ROR API (based on 11), GeoNames codes for intra-country regions
- **Author name disambiguation** based on Scopus author IDs (**98.3%** precision (no publication by others) and **90.6%** recall (all publications by X)) (12)

$$NMR_{i,t,k} = 1000 \times \frac{I_{i,t,k} - E_{i,t,k}}{N_{i,t}} \quad (1)$$

$$CMI_t = 100 \times \frac{\sum_i (I_{i,t,k} + E_{i,t,k})}{\sum_i N_{i,t}} \quad (2)$$

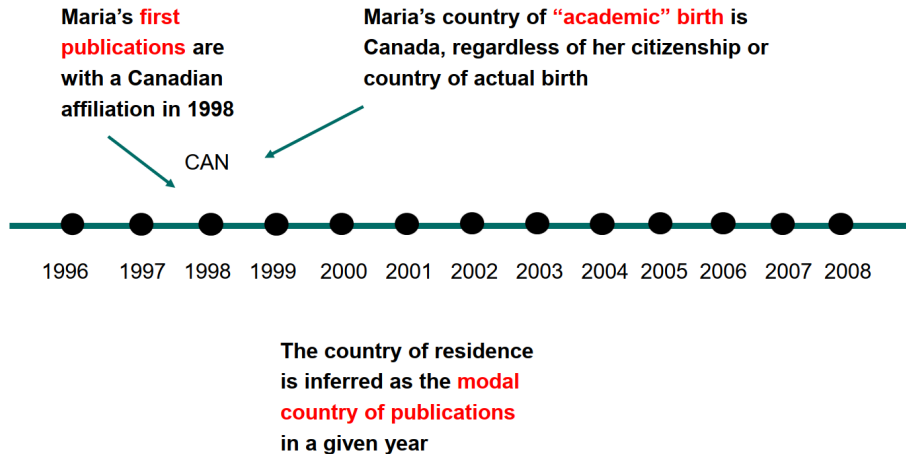
$$ANMR_t = 100 \times \frac{0.5 \sum_i |I_{i,t,k} - E_{i,t,k}|}{\sum_i N_{i,t}} \quad (3)$$

$$MEI_t = 100 \times \frac{\sum_i |I_{i,t,k} - E_{i,t,k}|}{\sum_i (I_{i,t,k} + E_{i,t,k})} \quad (4)$$

$$\text{Proportion Mobile}_i = 100 \times \sum_{p=1}^4 \frac{(I_{i,p,IL} - E_{i,p,IL}) + (I_{i,p,IT} - E_{i,p,IT})}{N_{i,p}} \quad (5)$$

i : sub-national region. t : year. p : six-years period. $N_{i,t}$: total number of scholars. Subscript k : type of data, internal (IL) or international (IT). $I_{i,t}$ inflow of scholars. $E_{i,t}$ outflow.

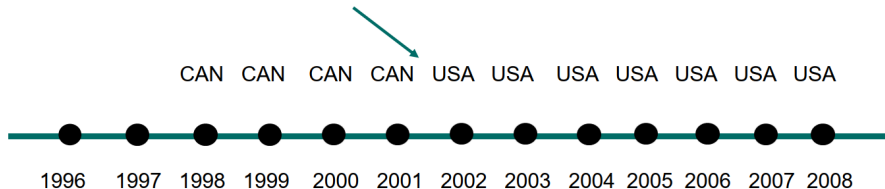
Identifying migration events (1/2)



Identifying migration events (2/2)

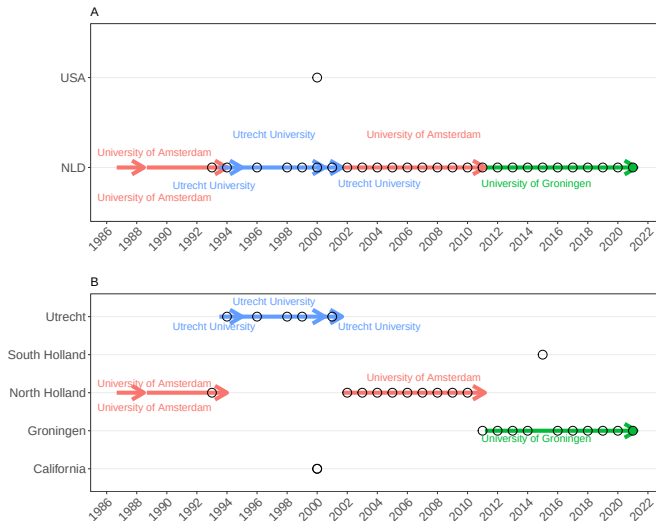


**Maria changes residence
from Canada to the US
between 2001 and 2002**



**Maria's modal countries
of publications over time**

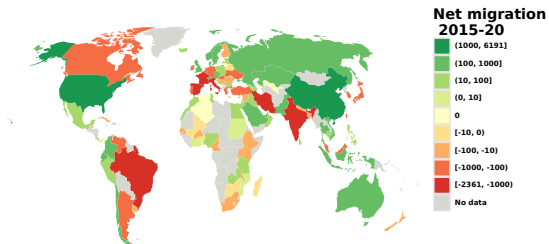
Example of integrated internal/international framework



Country level (bottom), sub-national (USA, top) migration



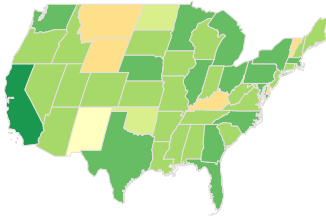
A



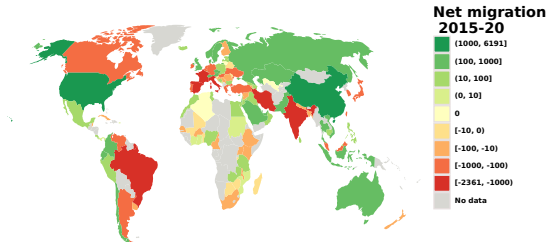
Country level (bottom), sub-national (USA, top) migration



B



A

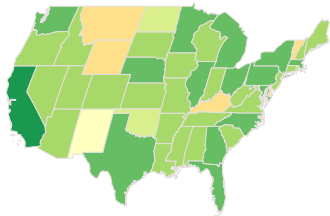


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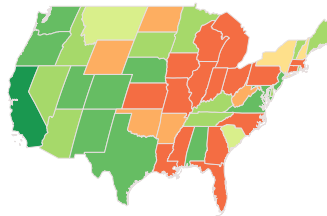


Mid-eastern states in the US lose scholars to western states.

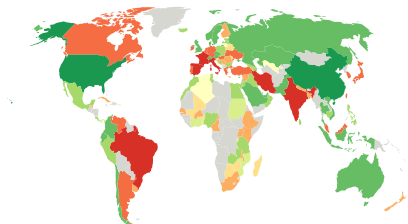
B



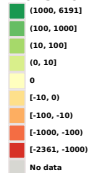
C



A



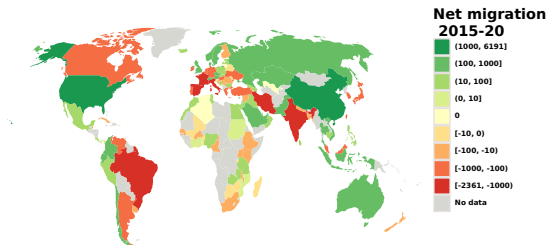
Net migration
2015-20



Country level (bottom), sub-national (DEU, top) migration



A

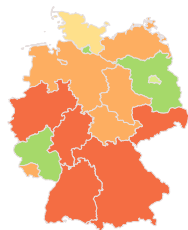


Country level (bottom), sub-national (DEU, top) migration

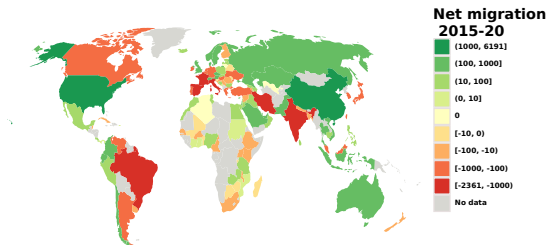


All four states receiving scholars internally (excluding Berlin/Hamburg), are sending internationally, i.e., *depopulation*

D



A



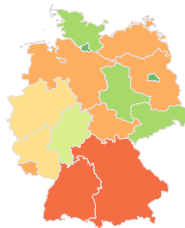
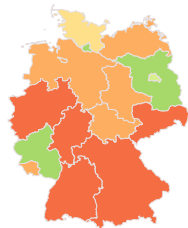
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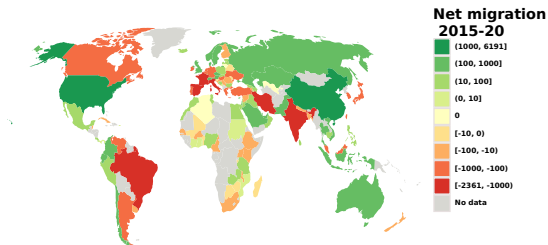
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D

E



A



Negative, positive or zero proportion of mobile scholars



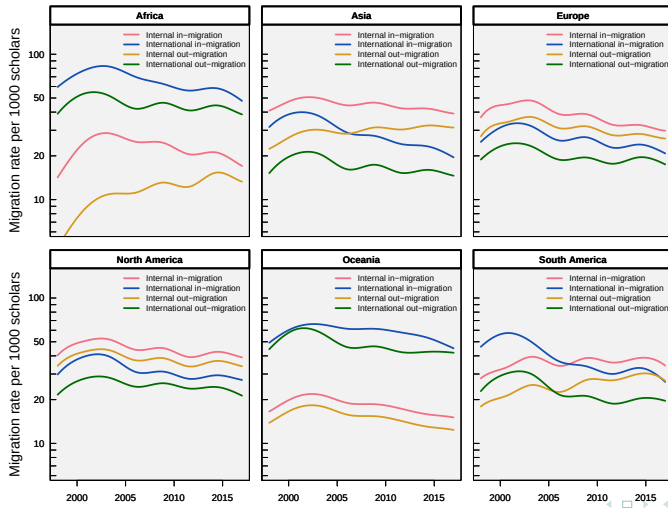
Continents/periods with **above 50%** of their regions **sending (bold red)** or **receiving (bold green)**.

Continent	Years	N negative	% negative	N positive	% positive	N zero	% zero	Sum N
Africa	1997-2002	17	54.8	14	45.2	0	0.0	31
	2003-2008	21	42.0	29	58.0	0	0.0	50
	2009-2014	35	51.5	33	48.5	0	0.0	68
	2015-2020	35	50.7	34	49.3	0	0.0	69
Asia	1997-2002	83	44.6	103	55.4	0	0.0	186
	2003-2008	98	43.4	126	55.8	2	0.9	226
	2009-2014	108	39.3	167	60.7	0	0.0	275
	2015-2020	121	43.4	158	56.6	0	0.0	279
Europe	1997-2002	108	53.5	94	46.5	0	0.0	202
	2003-2008	115	51.3	109	48.7	0	0.0	224
	2009-2014	122	47.5	135	52.5	0	0.0	257
	2015-2020	141	54.9	115	44.7	1	0.4	257
North America	1997-2002	30	40.0	45	60.0	0	0.0	75
	2003-2008	29	35.4	52	63.4	1	1.2	82
	2009-2014	40	47.6	44	52.4	0	0.0	84
	2015-2020	36	42.9	48	57.1	0	0.0	84
Oceania	1997-2002	10	62.5	6	37.5	0	0.0	16
	2003-2008	4	25.0	12	75.0	0	0.0	16
	2009-2014	5	31.2	11	68.8	0	0.0	16
	2015-2020	6	37.5	10	62.5	0	0.0	16
South America	1997-2002	12	33.3	24	66.7	0	0.0	36
	2003-2008	20	43.5	26	56.5	0	0.0	46
	2009-2014	22	44.0	27	54.0	1	2.0	50
	2015-2020	21	41.2	30	58.8	0	0.0	51

Change in immigration or outmigration



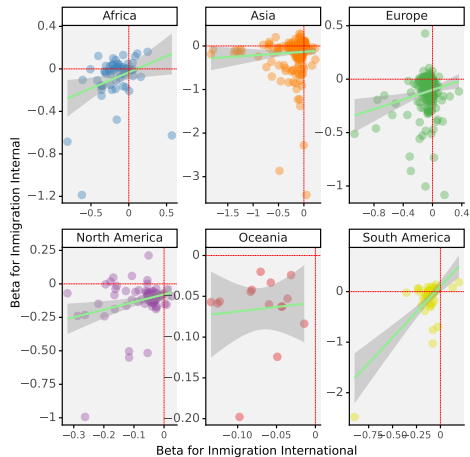
Straightforward picture in most continents where **in-migration** in internal or international dominates the respective trend in out-migration



Change in immigration or outmigration



Immigration Internal and International
(subnational regions worldwide)

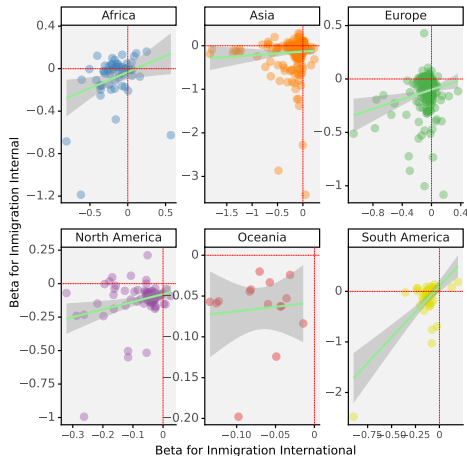


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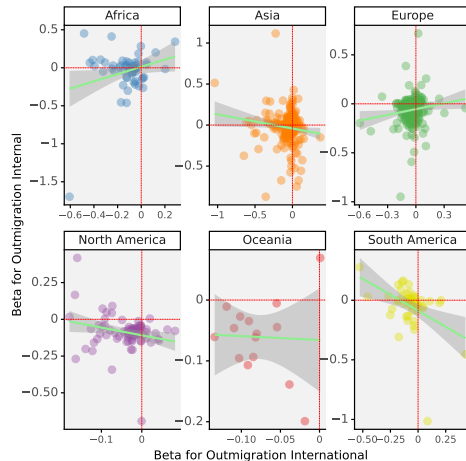


Straightforward picture (robust to size of threshold) in **immigration** (left) and **depopulation** in most continents in outmigration (right)

Immigration Internal and International
(subnational regions worldwide)



Outmigration Internal and International
(subnational regions worldwide)

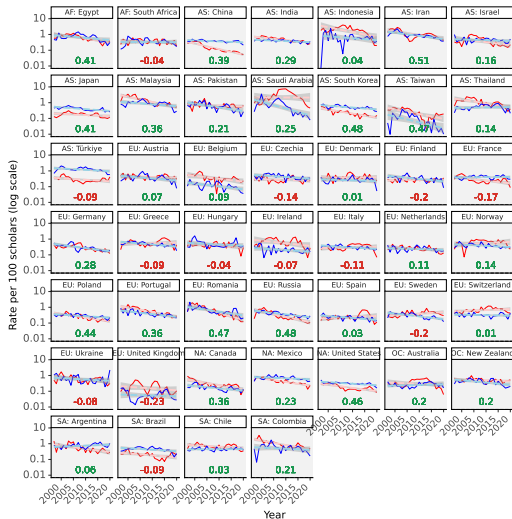


Trend of change in ANMR at country level



Most countries have **declining** or **stable** scholarly migration, and in some one of the two dominates the picture, e.g., China, Japan, Turkey, Ireland

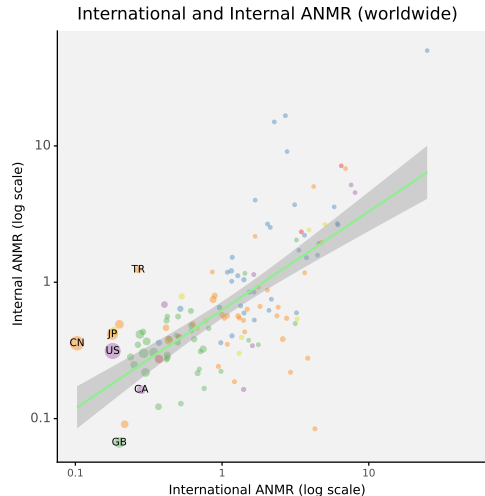
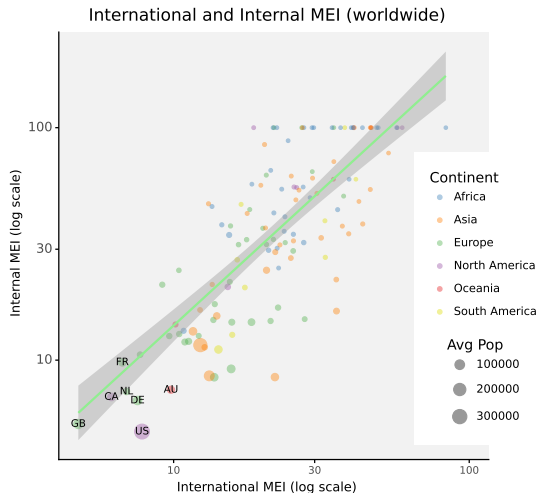
International and Internal ANMR (countries with +5k scholars per year)



International (X), Internal (Y) MEI/ANMR, country level



Exceptional cases: the USA or Australia where a much higher international MEI is coupled with a low internal MEI versus the UK or France



- ▶ Scopus published scholars (who publish actively and multiple times)
- ▶ Scopus coverage issues (over-representation of WEIRD countries, English language, hard sciences, etc.)
- ▶ Author's academic birth is not equivalent to their nationality and some of the outmigrations could be **return** migration.
- ▶ and many other limitations ...

Discussion and conclusions



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Discussion and conclusions



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- ▶ **The reverse**, being a popular sending origin to internal and international destinations is only true for African and European regions.
- ▶ In Asia, North/South America and Oceania the opposite trend is observed which could indicate **a depopulation effect**, where regions sending scholars to international destinations, in turn, receive scholars from internal origins (as cases of the US and Germany showed).



- ▶ Word2vec word embedding for multiple affiliations
- ▶ Identifying nationality using author's first and last name and trajectory of all affiliations to go closer to nationality (or families of geographical families, e.g., Spanish names, Asian names, ...)
- ▶ Comparison with employment history in ORCID as (hopefully) less biased towards WEIRD countries
- ▶ Gender, discipline, ..., differences

Thank you!



Questions and comments are welcome!

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Please tweet with hashtag
#BiblioDemography, a
tribute to James W. Vaupel
(1945-2022). Thanks to Ilya
and Jonas for bringing up
Jim's labeling idea!